◆ CMMD · CAPTURE INFRASTRUCTURE · SOLUTION

The Contributor Grid

Letting a user provide a live view of traffic on command — and turning that feed into something you own.

The kit to put in their hands, the architecture that taps it into the grid, and the provenance layer that makes it an asset.

CMMD-GRID-2026-0711

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DOCUMENT CONTROL

A Grade A1 solution for the contributor feed

DOCUMENT ID	CMMD-GRID-2026-0711
SERIES	The CZAR Empire — Capture Infrastructure
SUBJECT	Contributor-provided live traffic views: kit, ingest architecture, and provenance
ENTITY OF RECORD	CZAR Motor Mobility Data, Inc. (CMMD) · cross-references CZTVN Live Board
DATE	July 11, 2026
CHOSEN PATH	Open — 360 cam / phone app + your own media server (ownable)
DEPTH REGISTER	Strategy up front + build-spec detail
CLASSIFICATION	Internal record · Grade A1
AUTHOR	Josh Wayne — Founder / Architect

ON RIGHTS AND PRIVACY

This document specifies a system that ingests other people's camera feeds. The provenance section (V) is not optional garnish — consent, anonymization, and data rights are what make the feed lawful and ownable. Treat the legal specifics as a prompt to get proper agreements and counsel, not as legal advice.

HOW TO READ THIS SOLUTION

One problem, solved end to end

The goal is simple to say and specific to build: a contributor should be able to hand you a live view of traffic, on command, and that feed should tap cleanly into the grid you already built. This document answers the whole of it.

> *A live view of traffic, on command — that a user provides themselves.*
>
> THE OPERATOR, NAMING THE PROBLEM

CONTENTS

Six sections, with full-page dividers

THE CONTRIBUTOR GRID · SOLUTION · SECTION

I

The Choice

Open versus closed — and why ownable wins

OPEN
CHOSEN

OWN
NOT RENT

2
PATHS

I.1 The Fork

Two roads reach the same feature. The **closed** road leans on a consumer network — a Nexar or BlackVue dashcam whose live view runs through the maker's cloud and app. It is the fastest to field. The **open** road puts a 360 camera or a phone app in the contributor's hands and points it at *your own* media server. It is more to build — and it is the one this document chooses, for one reason.

I.2 The Reason Is the Whole Thesis

Access is not ownership. A closed network gives you a nice live view, but the stream, the terms, the recording rights, and the session limits are all theirs — you are renting the picture. The open path means the feed lands somewhere **you** control, on terms **you** set, in a form **you** can keep. For an enterprise whose entire moat is a proprietary, clean corridor record, that difference is not a detail. It is the point.

	OPEN PATH (CHOSEN)	CLOSED NETWORK
Where the feed lands	Your media server	Their cloud
Data rights	Yours, by contributor consent	Theirs, per their terms
Session limits	None you don't set	Often capped (e.g. ~3 min)
Recording / archive	Ownable capture	Rented, rarely exportable
Build effort	Higher	Lower
Fit to CMMD	Native	A shortcut, not the asset

> THE TRADE, STATED HONESTLY
>
> The open path costs more effort up front and asks a contributor to run a camera or an app rather than buy one box. In exchange, every feed it ingests is a candidate to **own** — which is the only kind of feed worth building a data company on.

THE CONTRIBUTOR GRID · SOLUTION · SECTION

II

The Kit

What to put in a contributor's hands, in tiers

3
TIERS

$0
ENTRY

RTMP
/ WEBRTC

II.1 Start Where Everyone Already Is

The cheapest contributor kit is a phone. A browser page using the camera, or a free RTMP broadcaster app, turns any phone into a live traffic feed pushed straight to your server — nothing to buy, nothing to mount. That is the entry tier, and it is how you get the first hundred contributors before anyone spends a dollar.

Tier 1 · Phone ENTRY

A web page (WebRTC) the contributor opens, or a free RTMP app (e.g. a Larix-style broadcaster). Zero cost, instant reach. Best for 'go live right now.'

Tier 2 · 360 camera IMMERSIVE

A Ricoh Theta X (~5.7K, wireless RTMP) or an Insta360 that pushes to a custom RTMP server. The immersive 'be there on the corridor' shot. ~$400–$800.

Tier 3 · Mounted rig NODE

A 360 cam or connected cam mounted and powered in-vehicle on its own hotspot/LTE — a semi-permanent contributor node for your most active partners.

II.2 Why 360 for the Middle Tier

A flat dashcam sees one lane of the story; a 360 camera captures the whole corridor at once, and on stream lets you pan the shot live. The open 360 cameras matter specifically because they push to a **custom** RTMP server — not just to YouTube — which is exactly the hook the grid needs. That single capability (custom server push) is the line between a contributor kit and a closed toy.

> *The kit is not the asset. The feed it sends to a server you own is.*

THE CONTRIBUTOR GRID · SOLUTION · SECTION

III

The Architecture

Contributor → media server → Live Board

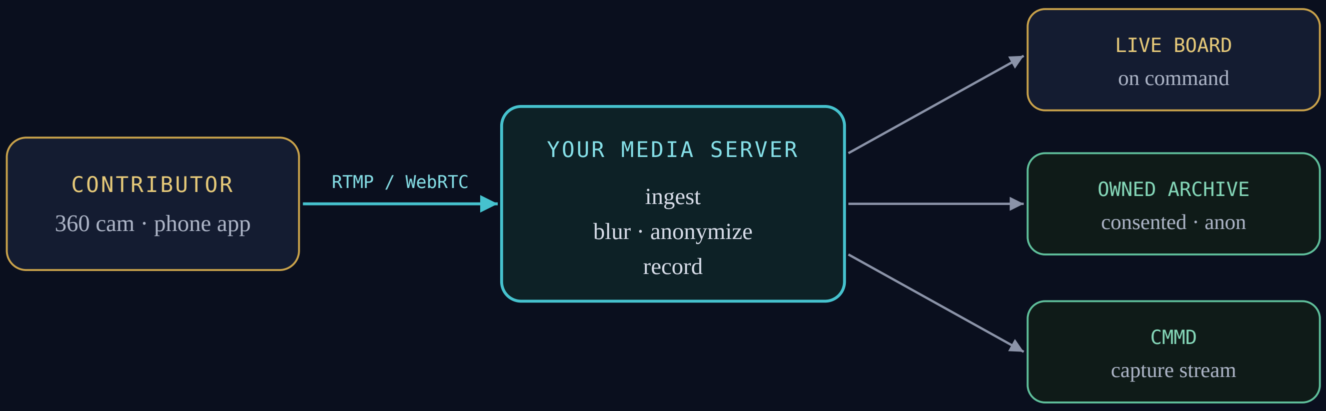
1
HUB

3
OUTPUTS

ON
COMMAND

III.1 The Whole Picture, One Diagram

A contributor's device pushes a live stream to a media server you run. That server is the hub: it ingests, anonymizes, records, and re-serves. From there the same feed fans out three ways — up to the Live Board on command, into the owned archive, and onward to CMMD as capture.



Nothing here is exotic. The contributor uses standard streaming (RTMP to publish, WebRTC for the low-latency 'on command' pull). The hub is one open-source server. The Live Board already has a camera slot waiting for it.

III.2 It Slots Into What You Already Built

The Live Board's city view already renders a camera as its hero, with a `cameraUrl` per corridor. A contributor feed is just another `cameraUrl` — the server hands the board a playback URL (HLS or WebRTC), and the board pulls it up exactly like a DOT camera. No new UI concept; a new *source* for an existing one.

III.3 What 'On Command' Really Means

Two flavors, and the doc specifies both. **HLS** is dead-simple and works everywhere, at a few seconds of delay — fine for a board that surfaces a contributor's corridor. **WebRTC** is the sub-second path — when you want to cut to a live contributor on air with no lag, that is the protocol. The build-spec wires HLS first (easy win) and WebRTC second (the broadcast finish).

> THE ONE DESIGN RULE
>
> Contributors **publish**; the grid **subscribes**. A contributor never connects to the Live Board directly — always through the server, which is where consent, anonymization, and recording are enforced. One hub, one place to hold the line.

THE CONTRIBUTOR GRID · SOLUTION · SECTION

IV

The Build-Spec

Server, stream keys, WebRTC, and the integration

1	HLS	KEY
OPEN-SRC SERVER	THEN WEBRTC	PER CONTRIBUTOR

IV.1 The Hub — One Open-Source Server

The whole hub can be a single open-source binary: **MediaMTX**. It ingests RTMP, re-serves the same stream as HLS, WebRTC, or RTSP, and can record to disk. It is the ownable equivalent of a streaming service — running on your box, on your terms. (For heavier WebRTC or scale, Ant Media or LiveKit are the upgrades; start with MediaMTX.)

```
# run the hub (one binary)
./mediamtx
# it opens: RTMP :1935 · HLS :8888 · WebRTC :8889
```

IV.2 A Key Per Contributor

Each contributor gets a unique stream path — that key is their identity and their on/off switch. They push to it; the grid reads from it.

```
# contributor publishes to:
rtmp://grid.yourdomain:1935/live/STREAMKEY

# the grid subscribes via HLS (simple):
http://grid.yourdomain:8888/STREAMKEY/index.m3u8

# or WebRTC / WHEP (sub-second):
http://grid.yourdomain:8889/STREAMKEY/whep
```

> WHY A KEY MATTERS BEYOND STREAMING
>
> The stream key is where consent attaches. No key is issued without a signed contributor agreement, so every feed on the grid is one you already have the right to use. Onboarding **is** the rights layer.

IV.3 Slotting Into the Live Board

The Live Board's hero already takes a `cameraUrl`. A DOT camera is a refreshing image; a contributor feed is a live stream, so the hero grows a `<video>` path — HLS via `hls.js` for the easy win, a WHEP player for the low-latency cut. The board lists whoever is live and pulls one up on command.

```
// Live Board: contributor feed as the hero (HLS)
const v = document.createElement('video'); v.autoplay = true; v.muted = true;
const hls = new Hls(); hls.loadSource(cor.cameraUrl); hls.attachMedia(v);
// cameraUrl = http://grid.yourdomain:8888/STREAMKEY/index.m3u8
```

IV.4 Recording — Only the Ownable

MediaMTX can record every stream to disk, but the grid records **only after** the anonymization step (Section V), and only what the contributor consented to. The recorded, blurred, consented file is the CMMD capture; the raw live pass-through, if unblurred, is shown but never stored.

```
# record path (gated on consent + blur):
record: yes
recordPath: /data/capture/STREAMKEY/%Y-%m-%d_%H-%M-%S
```

IV.5 Order of Operations

STEP	WHAT YOU STAND UP
1	MediaMTX on a server; open RTMP + HLS ports
2	Issue one test stream key; push from a phone app; confirm it plays
3	Point a Live Board hero at the HLS URL (add hls.js)
4	Add the consent-gated blur + record pipeline (Section V)
5	Add WebRTC / WHEP for the sub-second on-air cut

THE CONTRIBUTOR GRID · SOLUTION · SECTION

V

The Provenance Layer

Blur, consent, and clean data rights

BLUR
FIRST

OPT-IN
CONSENT

THE
MOAT

V.1 The Rule That Governs the Grid

Monitor the driving, not the driver — here, in a system that ingests other people's cameras, that doctrine becomes an engineering requirement, not a slogan. Before anything is **kept**, faces, license plates, and pedestrians are removed. The exterior flow of the corridor stays; the identities in it do not.

V.2 Where the Blur Happens

The gold standard is on-device blur before the frame ever leaves — the model a mature consumer network already uses (anonymize on-camera, opt-out available, delete-on-request honored). The open path can't always blur on-device, so the grid enforces it **server-side on ingest**: a detection-and-blur pass sits between the incoming stream and the recorder. A safe staging is to launch *live-view-only* (nothing stored) and switch on blur-plus-record once the pipeline is proven.

V.3 Consent Is the Contract

The contributor agreement — attached to the stream key — grants the right to use the **anonymized exterior** view, is strictly opt-in, and honors deletion on request. This is where to involve counsel; the specifics vary by state and use. The principle does not: nothing is kept that a contributor did not knowingly grant.

> THE NON-NEGOTIABLE
>
> Every ownable outcome in this document runs through clean provenance. A blurred, consented corridor feed is a licensable asset and a moat. The same feed unblurred or unconsented is a lawsuit and a brand wound. The discipline is not the cost of the opportunity — it **is** the opportunity.

> *Clean provenance is the product. The grid enforces it at one hub, or not at all.*

THE CONTRIBUTOR GRID · SOLUTION · SECTION

VI

The Payoff

The ownable capture stream into CMMD

FLYWHEEL
TWO-SIDED

5
NEXT MOVES

OWN
THE FEED

VI.1 A Two-Sided Flywheel

The grid spins in two directions at once. On the **media** side, contributor live views give CZTVN a well of on-command, on-the-ground footage no competitor has — which draws an audience, which draws more contributors. On the **data** side, every consented, anonymized feed is capture that compounds into CMMD's owned corridor record. The same act — a contributor going live — feeds both the voice and the asset.

THE LINE TO THE CAPSTONE

This is the contributor-scale version of the capstone's thesis: rented feeds evaporate; owned capture compounds. The Contributor Grid is one of the concrete taps through which CMMD's proprietary data actually begins to accumulate.

VI.2 Next Moves

ID	MOVE	DETAIL
G-01	Stand up the hub	MediaMTX on a server; RTMP in, HLS out; one test key end to end.
G-02	Prove the tap	Point a Live Board hero at a phone's HLS feed; see a contributor on the board.
G-03	Write the consent	Draft the contributor agreement (with counsel); bind it to key issuance.
G-04	Add the blur	Server-side anonymization on ingest; switch on gated recording.
G-05	Finish for air	WebRTC / WHEP for the sub-second on-command cut.

VI.3 The Operator, Verbatim

Preserved as spoken across the session that produced this solution.

> *"Is there any hardware, dash cam package, 360 camera, that has a way that it can tap into this grid — so that a user may, they themselves, provide a live view of traffic on command."*

> *"A document with a focus and solution to both of these is ideal for this."*

> *"Open path — 360 cam / phone app + your own server."*

END OF SOLUTION · CMMD-GRID-2026-0711 · GRADE A1
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